

Lab Session n. 3:
Solve the Single-Source Warehouse Location Problem
using a C++ framework for the *Backtracking* technique

Advanced Scheduling Systems

Prof. Andrea Schaerf

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Preliminaries

Consider the Single-Source Warehouse Location Problem. Download the available source files of Lab Session 3 in one folder.

Assignment 1

Design a backtracking strategy that processes the stores one at the time. Implement all the *hot spots* of the `BacktrackingOpt.hh` framework, namely `FirstOfLevel()`, `NextOfLevel()`, `Feasible()`, and `Cost()`, for the class `BacktrackingWLOpt.cc`.

In addition, (re)implement the two *cold spots* `GoUpOneLevel()` and `NonImprovingBranch()`. The first must unassign the store from the currently assigned warehouse. The second should prevent expanding a partial solution if it already costs at least as the current best solution.

Feel free to modify the classes `WL_Input` and `WL_Output` if necessary to improve the efficiency.

Note: the class `WL_Output` provides the possibility to unassign a warehouse to a store, by assigning the store to the dummy warehouse (-1); it also includes a data member (`total_cost`) that stores the cost of the current (partial) solution.

Assignment 2

Test the solution of Assignment 1 on the instances up to `wlp100`. Report for each instance the optimal value (if found within 3 minutes), otherwise report the best value found or the non-termination (if no solution found).

Send the final version of your work to `andrea.schaerf@uniud.it` (deadline: Sunday, May 3, 2026).